

Networks and Systems: MM10

Hans-Peter Schwefel

- **Mm1-7** Network Performance Analysis (L2-L5) + CAN/Flexray + clocks
- **Mm8-9** Discrete Event Simulation (HS)
- **Mm10** Dependability Modeling via Markov Chains (hps)
- **Mm11** Fault detection via Hidden Markov Chains (hps)

hps@es.aau.dk

Dependability

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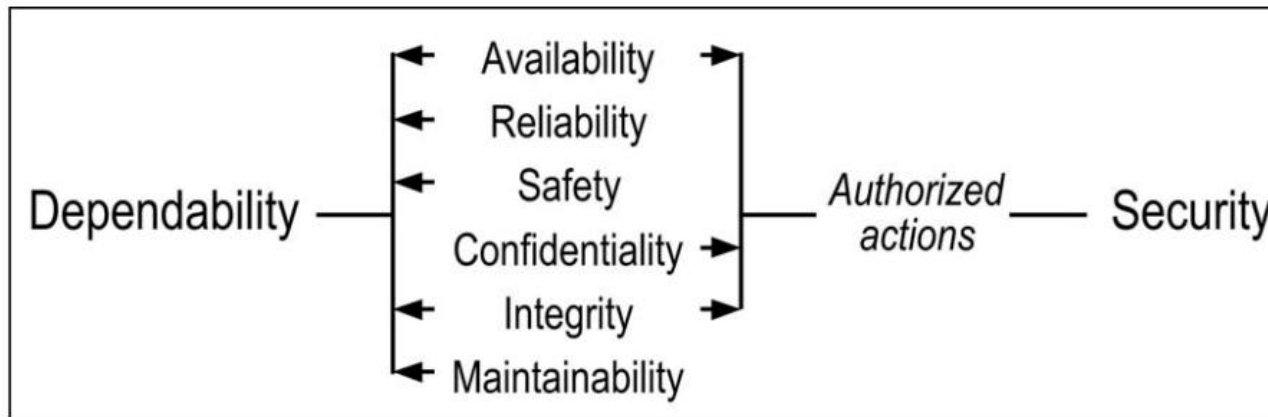
*the ability to avoid **service failures** that are more frequent and more severe than is **acceptable**.*

- Service Failure
 - ▣ System **fails to conform** to functional specification
- Error
 - ▣ **Deviation** from correct system state
 - ▣ Errors may lead to service failure (but not necessarily)
- Fault
 - ▣ The hypothesized or adjudged **cause** of an error (what caused the error?)

Attributes

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- **Availability:** readiness for correct service.
- **Reliability:** continuity of correct service.
- **Safety:** absence of catastrophic consequences on the user(s) and the environment.
- **Integrity:** absence of improper system alterations.
- **Maintainability:** ability to undergo modifications and repairs.



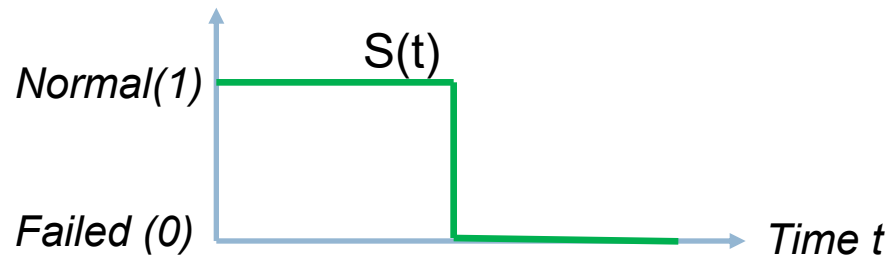
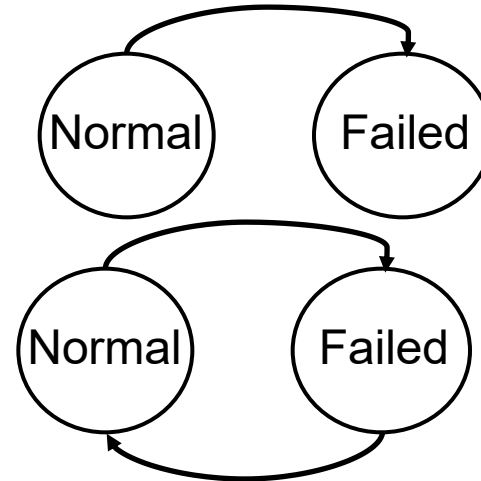
Binary System Model

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- System without repair

- System with repair

- Representation as stochastic process



- Availability: $A(t) := \Pr(\text{System in Normal State at Time } t) = \Pr(S(t)=1)$

- Reliability: $R(t) = \Pr(\text{System has been operational in time } [0, t]) = \Pr(S(t')=1 \text{ for all } 0 \leq t' \leq t)$

MM Learning outcomes

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- Represent faults and failures in Markov chain models
 - ▣ Understand product state space representation
- Be able to **numerically compute** availability, and reliability for a given system.
 - ▣ Understand how availability and reliability models differ
- Understand Markov chain models for reliability/availability analysis of multi-compoment systems

Plan

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1. Markov models revisited
using the single component example
2. Reliability/availability models
3. ... of multiple components with dependencies
4. ... multiple fault types
5. ... non-constant failure rates
6. Exercises: TMR example and more

Revision: Exponential Distributions

Exponentially distributed RV

- Single parameter: rate λ
- Density function $f(x) = \lambda \exp(-\lambda x)$, $x > 0$
- Cdf: $F(x) = 1 - \exp(-\lambda x)$, Reliability function: $R(x) = \exp(-\lambda x)$
- Moments: $E\{X\} = 1/\lambda$; $\text{Var}\{X\} = 1/\lambda^2$, $C^2 = \text{Var}\{X\} / [E\{X\}]^2 = 1$

Important properties:

- Memory-less: $\Pr(X > x+y \mid X > x) = \exp(-\lambda y)$
- Properties of two independent exponential RV: X with rate λ , Y with rate μ
 - ▣ Distribution of $\min(X, Y)$: exponential with rate $(\lambda + \mu)$
 - ▣ $\Pr(X < Y) = \lambda / (\lambda + \mu)$

Revision: Markov Processes: Discrete Time

- **State-Space:** finite or countable infinite, w/o.l.g. $E=\{0,1,2,\dots,K\}$ ($K=\infty$ also allowed)
- **Transition probabilities:** $p_{jk}=\text{Pr}(\text{transition from state } j \text{ to state } k)$
- X_i = RV indicating the state that the Markov process is in in step i
- **'Markov Property':** State in step i only depends on state in step $i-1$
 - ▣ $\text{Pr}(X_i=k)=\sum_j [\text{Pr}(X_{i-1}=j)*p_{jk}]$
 - ▣ Consequence: geometrically distributed state-holding times
- **Computation of state probabilities**
 - ▣ Probability vector $\underline{\pi}_i$
 - ▣ Matrix notation for behavior of $\underline{\pi}_i$: $\underline{\pi}_i = \underline{\pi}_{i-1} \underline{P}$
 - ▣ Steady-state solution: $\underline{\pi} = \underline{\pi} \underline{P}$
- Assumption here: irreducibility (no transient states), non-periodicity, and homogeneity

Revision: Continuous Time Markov Processes

- Defined by
 - ▣ **State-Space:** finite or countable infinite, w/o.l.g. $E=\{0,1,2,\dots,K\}$ ($K=\infty$ also allowed)
 - ▣ **Transition rates:** μ_{jk}
 - Holding time in state j : exponential with rate $\sum_{k \neq j} \mu_{jk}$
 - Transition probability from state j to k : $\mu_{jk} / \sum_{l \neq j} \mu_{jl} =: \mu_{jk} / \mu_j$
- X_t = RV indicating the current state at time t ; $\pi_i(t) := \Pr(X_t = i)$
- 'Markov Property': transitions do not depend on history but only on current state
 $t_0 < t_1 < \dots < t_n, \forall i_0, i_1, \dots, i_n, j \in E$
$$\frac{P\{X(t_{n+1}) = j \mid X(t_n) = i_n, \dots, X(t_0) = i_0\}}{P\{X(t_{n+1}) = j \mid X(t_n) = i_n\}} = 1$$
- Computation of steady-state probabilities
 - ▣ Chapman Kolmogorov Equations: $d\pi_i(t)/dt = -\mu_i \pi_i(t) + \sum_{j \neq i} \mu_{ji} \pi_j(t)$
 - ▣ Flow-balance equations, steady-state ($t \rightarrow \infty$): $\mu_i \pi_i = \sum_{j \neq i} \mu_{ji} \pi_j$
- Here: restriction to irreducible, homogeneous processes

Focus this lecture: Transient state probabilities of Continuous time Markov Chains

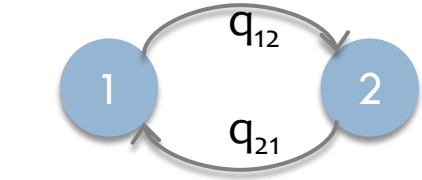
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□ Definitions:

- State-space: finite or countable infinite

E.g., $E = \{1, 2, \dots, N\}$ ($N = \infty$ also allowed)

- Transition rate matrix: $\underline{Q}$ is $n \times n$ (n is number of states)



$$\underline{Q} = \begin{bmatrix} -q_{12} & q_{12} \\ q_{21} & -q_{21} \end{bmatrix}$$

- q_{ij} is transition rate (events/time) between states i and j , where $q_{ij} > 0$.

- Q 's diagonal entries are negative so that rates sum to zero: $\sum_{all\ j} q_{ij} = 0$

- For homogeneous MC: $\underline{Q}(t) = \underline{Q}$.

- State holding time (or inter-event time): $V(t)$ is exponentially distributed

- cdf of state holding time for state i : $P[V(i) \leq t] = 1 - e^{-\Lambda(i)t}$, $t \geq 0$.

- $\Lambda(i)$ is rate of exponential state holding time distribution

- Computation of state probabilities:

- State probabilities at time t : $\underline{P}(t) = \underline{P}(0) \expm(\underline{Q}t)$,
($\underline{P}(0)$ is vector with initial state probabilities)

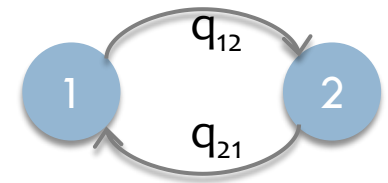
- Steady state solution: as $t \rightarrow \infty$, we expect $d\pi/dt \rightarrow 0$ (for irreducible and positive recurrent MC)
(the change in $\underline{P}(t)$) should settle for large values of t
 $\underline{\pi} \underline{Q} = 0$ and $\underline{\pi}(1, \dots, 1)' = 1$

Markov Model: Single component (with or without) repair

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□ Single component

- Exponential time to failure (e.g. with mean 10)
- Exponential time to repair (e.g. with mean 1)



□ Constructing the model

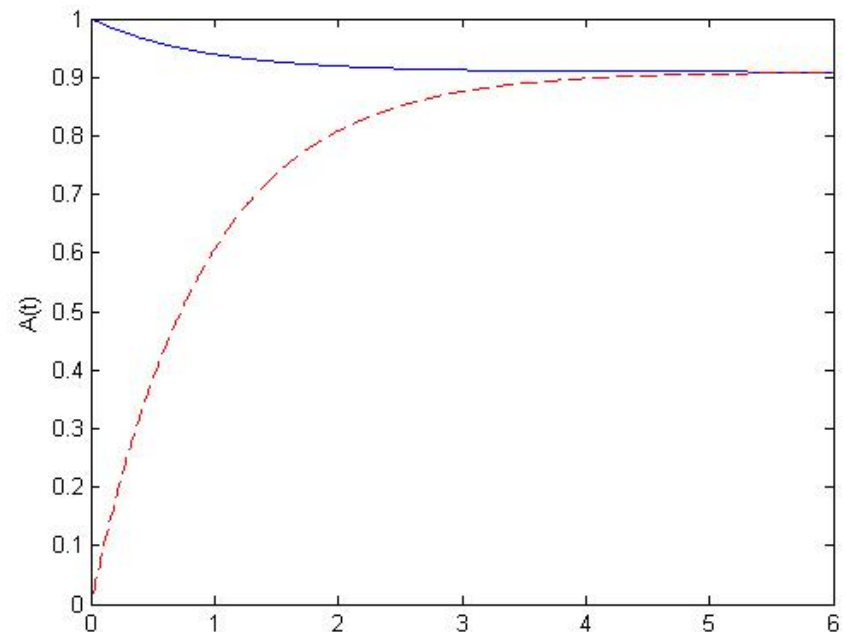
- Resulting stationary availability = ?
- Transition rates in Markov Chain
 - $\underline{Q}=?$

$$\underline{Q} = \begin{bmatrix} -q_{12} & q_{12} \\ q_{21} & -q_{21} \end{bmatrix}$$

□ Transient probabilities

$$\underline{p}(t) = \underline{p}(0) \expm(\underline{Q}t) (1,0)'$$

- For $\underline{p}(0) = [1,0]$
- For $\underline{p}(0) = [0,1]$
- For $\underline{p}(0) = [10/11, 1/11]$?



Multiple components

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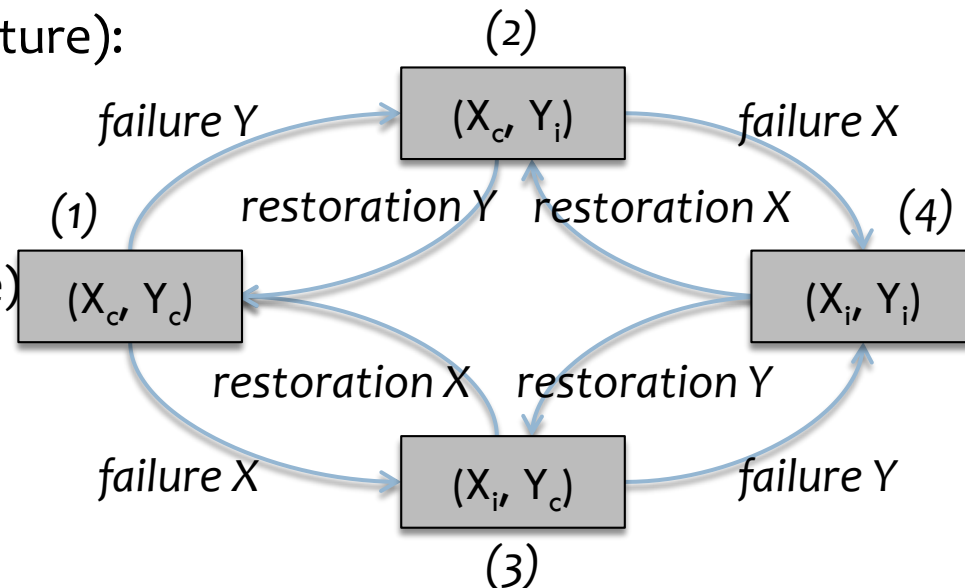
- Example – Product State Spaces:
 - ▣ System: two components X,Y; 1 repairman each
 - ▣ Component states:
 - X_c, Y_c (correct service); X_i, Y_i (incorrect service)
 - System states: (X_c, Y_c) , (X_i, Y_c) , (X_c, Y_i) , (X_i, Y_i)

Non-redundant system (serial structure):

- Correct service: (1)
- Incorrect service: (2), (3), (4)

Redundant system (parallel structure)

- Correct service: (1), (2), (3)
- Incorrect service: (4)

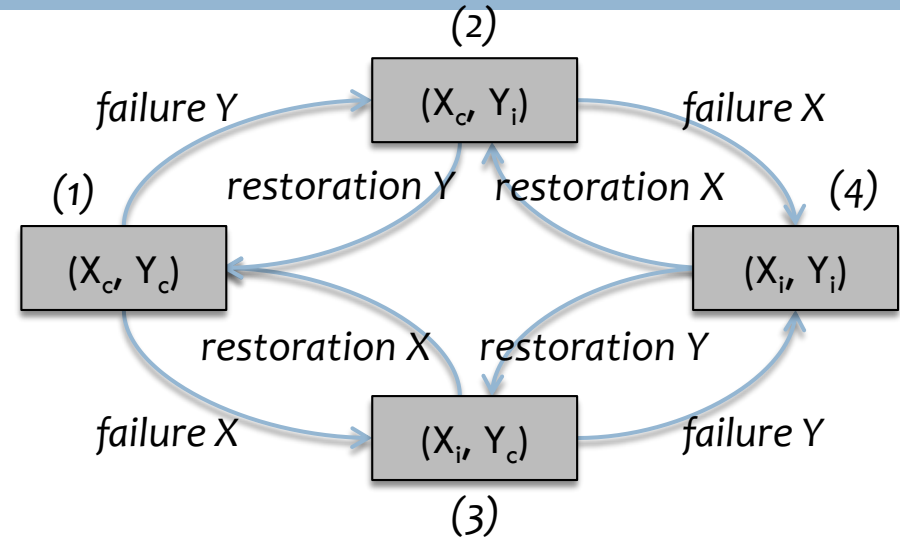
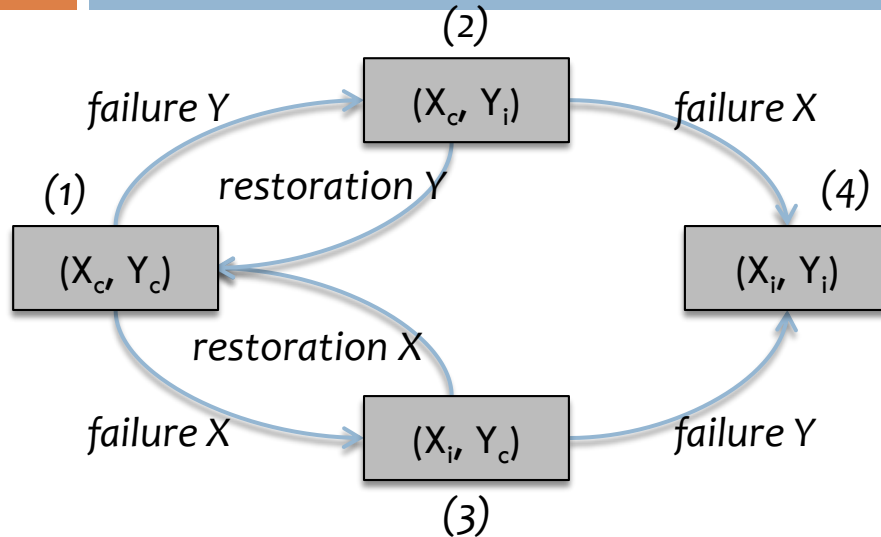


Redundant System (Parallel Structure)

Reliability model

Availability model

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$$R(t) = P_1(t, \underline{Q}_{\text{red}}) + P_2(t, \underline{Q}_{\text{red}}) + P_3(t, \underline{Q}_{\text{red}}), \quad A(t) = P_1(t, \underline{Q}) + P_2(t, \underline{Q}) + P_3(t, \underline{Q})$$

$P_k(t)$: probability of system in state k at time t

- Computation of $P_j(t)$ depends on the probability distributions associated to state transitions
- Homogeneous Markov Chains: constant transition rates.

Availability computation

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Matrix form:

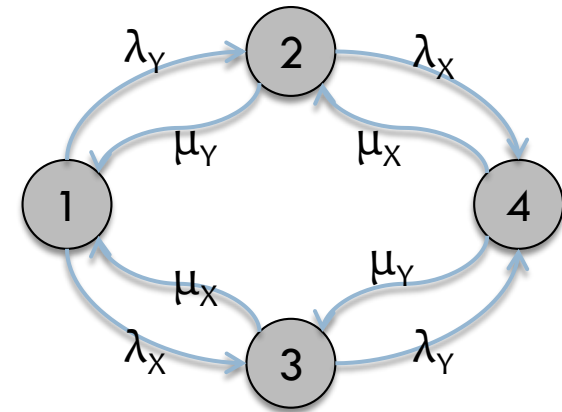
$$\square \frac{d\underline{P}(t)}{dt} = \underline{P}(t) \underline{Q}$$

- $\underline{P}(t)$: state probability vector
- $\underline{Q}$: infinitesimal generator matrix (transition rate matrix)

$$\square \text{Solution: } \underline{P}(t) = \underline{P}(0) \exp(\underline{Q}t)$$

- (use matlab's `expm()` command for matrix exponential.)

$$\underline{Q} = \begin{pmatrix} -(\lambda_X + \lambda_Y) & \lambda_Y & \lambda_X & 0 \\ \mu_Y & -(\lambda_X + \mu_Y) & 0 & \lambda_X \\ \mu_X & 0 & -(\mu_X + \lambda_Y) & \lambda_Y \\ 0 & \mu_X & \mu_Y & -(\mu_X + \mu_Y) \end{pmatrix}$$



$$A(t) = P_1(t) + P_2(t) + P_3(t) = \underline{P}(t) \underline{1}_3^A$$

$$\underline{1}_3^A = \begin{pmatrix} 1 & 1 & 1 & 0 \end{pmatrix}^T : \text{summation vector}$$

Reliability computation

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□ Matrix form:

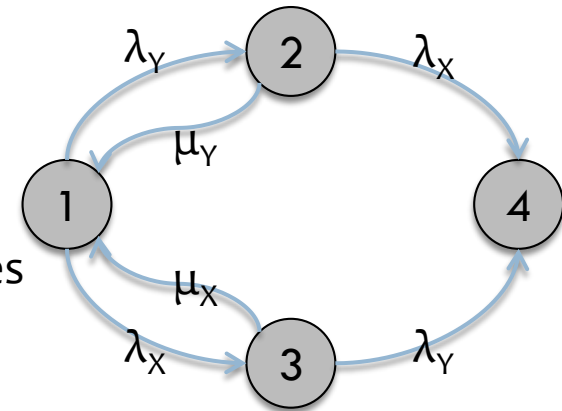
□ $\frac{d\underline{P}(t)}{dt} = \underline{P}(t) \underline{Q}_{red}$

■ $\underline{P}(t)$: state probability vector

■ $\underline{Q}_{red}$: transition rate matrix – correct service states

□ Solution: $\underline{P}(t) = \underline{P}(0) \expm(\underline{Q}_{red}t)$

■ (use matlab's `expm()` command for matrix exponential.)



$$A(t) = P_1(t) + P_2(t) + P_3(t) = P(t) \underline{1}_3^A$$

: summation vector

$$\underline{1}_3^A = \begin{pmatrix} 1 & 1 & 1 & 0 \end{pmatrix}^T$$

Model variants: some frequent cases

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Q: what modifications the Markov chain result in the following cases?

□ **Redundancy types:** Assume the redundant system

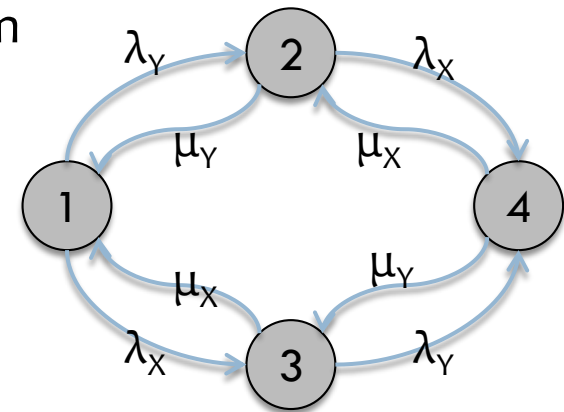
□ Assume that only active components fail (e.g. SW)

■ In case of active stand-by

■ In case of activation on demand

■ For instantaneous activation

■ If activation of the redundancy takes an exponential time period of mean duration D



□ **Correlated faults/repairs**

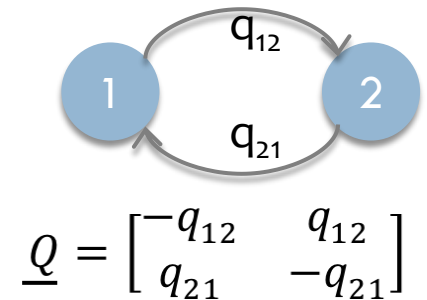
□ Assume that if one component fails, the failure rate of the second one increases while that component is failed

□ The speed of repair degrades by 50% if both components are down

Modelling multiple fault-types

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- Back to Single component with exponential assumptions
- Modification: multiple, different fault types, e.g.
 - ▣ Hardware error, mTTF = 100, mTTR = 10 (replace part);
 - ▣ Software Error, mTTF = 10, mTTR=1 (reboot)
 - ▣ Etc.
- Constructing the model
 - ▣ State-space for reliability? State-space for availability?
 - ▣ Transition rates?
 - ▣ Impact on availability and reliability?
- Modelling correlation of faults, e.g.
 - ▣ Frequent reboots increase the probability of hardware faults



Modeling non-constant failure rates

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- So far we assumed exponential failure (and repair) times
- In many fault cases hazard rates are not constant
 - ▣ Example: bath-tub like shapes due to production errors and aging
- How to represent this in Markov models?

non-constant failure rates (cntd)

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□ Intuitive approach

▣ Keep track of time in different states

- State 1: high failure rate (production error)
- State 2: low failure rate (intermediate phase)
- State 3: high failure rate (wear-out)

Different notions of repair

□ Distribution fitting

- ▣ Calculate $R(t)$ or $f(t)$ from failure rate
- ▣ Fit phase-type distribution (beyond scope of this lecture)

Example: TMR Systems

(Triple Module Redundancy)

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- “2-out-of-3” system with perfect voter

$$R(t) = 3 (R_c(t))^2 - 2 (R_c(t))^3$$

Single component:

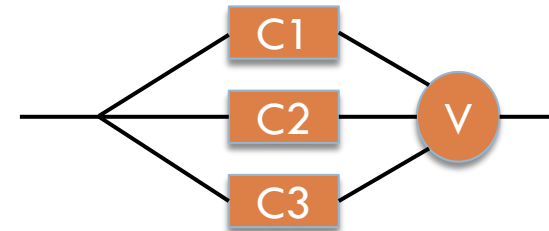
$$R_c(t) = \exp(-\lambda t) \quad \text{vs.}$$

$$MTTF_c = 1 / \lambda \quad >$$

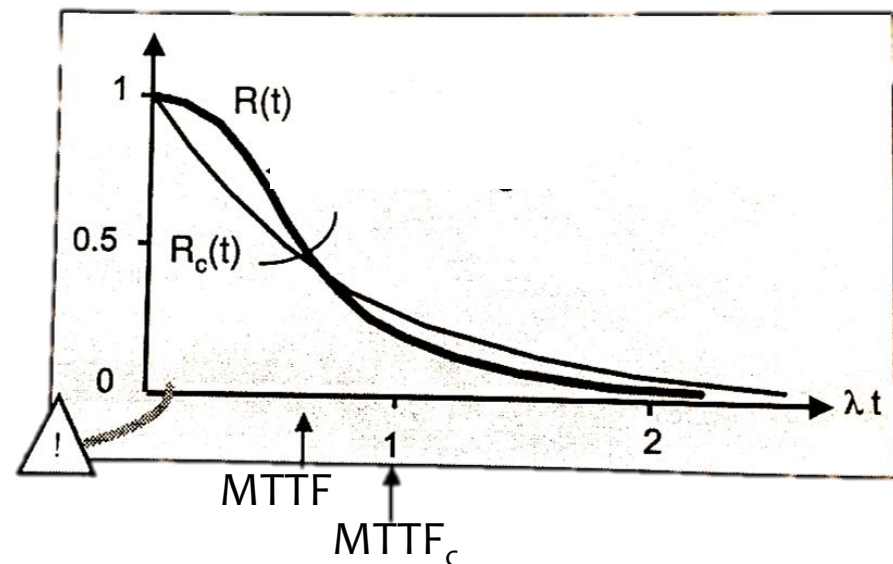
TMR System:

$$R(t) = 3 (\exp(-\lambda t))^2 - 2 (\exp(-\lambda t))^3$$

$$MTTF = 5 / (6 \lambda)$$



- TMR has lower MTTF but higher $R(t)$ in short term. It is useful for short mission times.
- Be careful when using MTTF to characterize dependability, as it does not describe the distribution!



Summary

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1. Markov models revisited
2. Reliability/availabiltiy models
3. ... of multiple components with dependencies
4. ... multiple fault types
5. ... non-constant failure rates
6. TMR example
7. Summary

Outlook

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- ,Manual‘ construction of Markov chains may be tedious
→ Higher-layer abstractions, e.g. Petri-Nets, Stochastic Activity Networks
(which can be translated into Markov chains)
- General distributions → Phase-type distributions
- Markov models for fault-detection/diagnosis → Hidden Markov models (see later lecture)
- Highly available components (rare failure events) → numerical methods for computation of transient and steady-state probabilities

Exercises

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1. Multiple components

- Plot the reliability and availability for the redundant two-component state based Markov model using failure rates $\lambda_X=0.001$ and $\lambda_Y=0.002$ and a common restoration rate of $\mu_X=\mu_Y=0.05$ as a function of time t .
- Plot the same curves for the model where only active components fail with instantaneous activation.
- Do the same for the non-redundant model and compare the reliability/availability results.
- Assume that speed of repair is reduced to 70% in case of both components down; plot the corresponding availability

Exercises

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2. Fault models: given is a single component that is subject to three types of faults with the following parameters: $mttF_1=10$, $mttR_1=1$; $mttF_2=50$, $mttR_2=10$; $mttF_3=100$; $mttR_3=10$;

- ▣ Calculate and plot the reliability and availability function for this component
- ▣ Assume that faults of the other type can also occur while repair is going on. Modify your model and plot the resulting $A(t)$

3. Take the single component model from slide 11 ($mttf=10$). The $mttf$ is now replaced by a two exponential phases, where in the first phase of mean duration D $mttf=100$ and after which the $mttF$ reduces to 5. Draw the corresponding Markov chain model, implement it in Matlab and calculate $R(t)$ for different values of $D=1,5,10,100$.

References

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- *Basic Concepts and Taxonomy of Dependable and Secure Computing*, A Avizienis, JC Laprie, B Randell, C Landwehr, Dependable and Secure Computing, IEEE Transactions on, 2004.
- Presentation: “Resilience evaluation with regard to accidental and malicious threats”, Mohamed Kaâniche, RESIST summer school, 24-28 September, Porquerolles, France.

Markov Chains: see references to C&L from MM1